

Havelock Lagrangian Eigenvalues: First-Principles Derivation

Sandbox for Paper I, Item #1

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1 Setup

Let $z_k = e^{i\theta_k}$, $\theta_k = 2\pi k/N$, $k = 0, \dots, N-1$, be the regular N -gon of unit-circulation point vortices on the unit circle. The Thomson energy is

$$H = - \sum_{j < k} \ln |z_j - z_k|. \quad (1)$$

We perturb each vortex in its *local* (radial, tangential) frame:

$$z_k \longmapsto z_k(1 + \delta r_k) + i z_k \delta t_k, \quad (2)$$

with $\delta r_k, \delta t_k \in \mathbb{R}$ small. The $\mathbb{Z}_N$ cyclic symmetry $k \mapsto k+1 \pmod{N}$ permutes the vortices and commutes with $\nabla^2 H$, so the Hessian block-diagonalizes in the discrete Fourier basis. The mode- m cosine sector is parametrized by

$$\delta r_k = a \cos(2\pi m k/N), \quad \delta t_k = b \cos(2\pi m k/N), \quad m \in \{1, \dots, N-1\}, \quad (3)$$

(modes m and $N-m$ yield identical blocks — the palindromic pair). The corresponding 2×2 Hessian block in the (a, b) basis is what we compute.

2 Two auxiliary identities

Two closed-form trigonometric sums enter the block computation and are recorded here. Both are verified to 30 decimal digits in `numerical_check.py` (`check_havelock_identity_rr`, `check_cosecant_squared_sum`).

Lemma 1 (Havelock identity). *For all integers $N \geq 2$ and $m \in \{1, \dots, N-1\}$,*

$$S(m) := \sum_{p=1}^{N-1} \frac{1 - \cos(2\pi m p/N)}{4 \sin^2(\pi p/N)} = \frac{m(N-m)}{2}. \quad (4)$$

Proof. Using $1 - \cos(2\pi m p/N) = 2 \sin^2(\pi m p/N)$,

$$S(m) = \sum_{p=1}^{N-1} \frac{\sin^2(\pi m p/N)}{2 \sin^2(\pi p/N)}.$$

The Dirichlet-kernel identity $\sum_{p=1}^{N-1} \sin^2(\pi m p/N) / \sin^2(\pi p/N) = m(N-m)$ for $1 \leq m \leq N-1$ (provable via the Chebyshev expansion $\sin(mx)/\sin(x) = U_{m-1}(\cos x)$ and orthogonality of the U -polynomials; see e.g. Gradshteyn and Ryzhik, Formula 1.342) yields the claim. $\square$

Lemma 2 (Cosecant-squared sum). *For $N \geq 2$,*

$$\sum_{p=1}^{N-1} \frac{1}{4 \sin^2(\pi p/N)} = \frac{N^2 - 1}{12}. \quad (5)$$

Proof. Standard; follows from partial-fraction expansion of $\cot(Nz)$ and taking the limit $z \rightarrow 0$, or equivalently from differentiating $\prod_{p=1}^{N-1} \sin(\pi p/N) = N/2^{N-1}$. $\square$

Corollary 1 (Cosine-weighted inverse-sine sum). *For $N \geq 2$ and any integer k ,*

$$C(k) := \sum_{p=1}^{N-1} \frac{\cos(2\pi kp/N)}{4 \sin^2(\pi p/N)} = \frac{N^2 - 1}{12} - \frac{k(N - k)}{2} \quad \text{for } 0 \leq k \leq N. \quad (6)$$

The expression is periodic in k modulo N , i.e. $C(k + N) = C(k)$.

Proof. Subtract (4) from (5): $C(k) = (N^2 - 1)/12 - S(k) = (N^2 - 1)/12 - k(N - k)/2$. Periodicity is immediate from the integrand. $\square$

3 Radial-radial block

Under a purely radial mode- m perturbation $\delta r_k = a \alpha_k$ with $\alpha_k = \cos(2\pi mk/N)$ (and $b = 0$ in (3)), the squared pair distance expands as

$$|z_j(a) - z_k(a)|^2 = |z_j - z_k|^2 [1 + a(\alpha_j + \alpha_k)] + a^2 Q_{jk} + O(a^3), \quad (7)$$

with

$$Q_{jk} := \alpha_j^2 + \alpha_k^2 - 2\alpha_j \alpha_k \cos(\theta_j - \theta_k), \quad |z_j - z_k|^2 = 4 \sin^2(\pi(j - k)/N).$$

(To see this, write $z_k(a) = z_k(1 + a\alpha_k)$, compute $|z_j(a) - z_k(a)|^2 = |z_j - z_k|^2 + 2a \operatorname{Re}[(z_j - z_k)(\alpha_j z_j - \alpha_k z_k)] + a^2 |\alpha_j z_j - \alpha_k z_k|^2$, and simplify the real part using $\operatorname{Re}[(e^{-i\theta_j} - e^{-i\theta_k})(\alpha_j e^{i\theta_j} - \alpha_k e^{i\theta_k})] = (\alpha_j + \alpha_k)(1 - \cos(\theta_j - \theta_k))$.)

Expanding $-\ln |z_j(a) - z_k(a)| = -\frac{1}{2} \ln(|z_j(a) - z_k(a)|^2)$ to second order in a via $\ln(1 + x) = x - x^2/2 + O(x^3)$,

$$-\ln |z_j(a) - z_k(a)| = -\ln |z_j - z_k| - \frac{a}{2}(\alpha_j + \alpha_k) + a^2 \left[\frac{(\alpha_j + \alpha_k)^2}{4} - \frac{Q_{jk}}{2|z_j - z_k|^2} \right] + O(a^3). \quad (8)$$

The linear term, summed over pairs $j < k$, gives $-(a/2)(N-1) \sum_k \alpha_k = 0$ since $\sum_k \cos(2\pi mk/N) = 0$ for $m \in \{1, \dots, N-1\}$ (equilibrium).

Write β_m for the coefficient of a^2 in $H(a) - H(0)$:

$$\beta_m = \sum_{j < k} \left[\frac{(\alpha_j + \alpha_k)^2}{4} - \frac{Q_{jk}}{2|z_j - z_k|^2} \right].$$

The unit-normalized radial Fourier mode vector $\hat{e}_r$ (in the $2N$ -dimensional Cartesian basis) has squared norm $\|\hat{e}_r\|^2 = \sum_k \alpha_k^2 = N/2$ (valid for $1 \leq m \leq N-1$, $m \neq N/2$; the case $m = N/2$ is handled symmetrically and gives the same final formula by continuity). The Hessian entry in this normalized direction is

$$(\nabla^2 H)_{rr}^{(m)} = \frac{2\beta_m}{\|\hat{e}_r\|^2} = \frac{4\beta_m}{N} = \frac{2}{N} \sum_{j \neq k} \left[\frac{(\alpha_j + \alpha_k)^2}{4} - \frac{Q_{jk}}{2|z_j - z_k|^2} \right]. \quad (9)$$

(The factor 2 comes from $H''(0) = 2 \cdot (\text{coefficient of } a^2)$; the factor 2 converting $\sum_{j < k}$ to $\sum_{j \neq k}$ cancels half of it.)

Parametrize pairs by $p = k - j \pmod{N}$ with $p \in \{1, \dots, N-1\}$ and apply the N -fold cyclic symmetry. Two elementary sums are needed:

$$\sum_{j=0}^{N-1} \alpha_j^2 = \frac{N}{2}, \quad \sum_{j=0}^{N-1} \alpha_j \alpha_{j+p} = \frac{N}{2} \cos(2\pi mp/N).$$

From these,

$$\sum_{j=0}^{N-1} (\alpha_j + \alpha_{j+p})^2 = N(1 + \cos(2\pi mp/N)), \quad \sum_{j=0}^{N-1} Q_{j,j+p} = N(1 - \cos(2\pi mp/N) \cos(2\pi p/N)).$$

Substituting into (9),

$$(\nabla^2 H)_{rr}^{(m)} = \sum_{p=1}^{N-1} \frac{1 + \cos(2\pi mp/N)}{2} - \sum_{p=1}^{N-1} \frac{1 - \cos(2\pi mp/N) \cos(2\pi p/N)}{4 \sin^2(\pi p/N)}. \quad (10)$$

The first sum is elementary: $\sum_{p=1}^{N-1} (1 + \cos(2\pi mp/N))/2 = (N-1)/2 - 1/2 = (N-2)/2$, since $\sum_{p=1}^{N-1} \cos(2\pi mp/N) = -1$ for $m \in \{1, \dots, N-1\}$.

The second sum reduces via the product-to-sum identity $\cos A \cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$:

$$1 - \cos(2\pi mp/N) \cos(2\pi p/N) = 1 - \frac{1}{2} \cos(2\pi(m-1)p/N) - \frac{1}{2} \cos(2\pi(m+1)p/N).$$

Applying Lemma 2 and Corollary 1,

$$\begin{aligned} \sum_{p=1}^{N-1} \frac{1 - \cos(2\pi mp/N) \cos(2\pi p/N)}{4 \sin^2(\pi p/N)} &= \frac{N^2 - 1}{12} - \frac{1}{2} \left[\frac{N^2 - 1}{12} - \frac{(m-1)(N-m+1)}{2} \right] \\ &\quad - \frac{1}{2} \left[\frac{N^2 - 1}{12} - \frac{(m+1)(N-m-1)}{2} \right] \\ &= \frac{(m-1)(N-m+1) + (m+1)(N-m-1)}{4} = \frac{m(N-m) - 1}{2}. \end{aligned}$$

Combining,

$$(\nabla^2 H)_{rr}^{(m)} = \frac{N-2}{2} - \frac{m(N-m) - 1}{2} = \frac{N-1-m(N-m)}{2}. \quad (11)$$

This is the radial-radial block eigenvalue of the *unshifted* energy Hessian $\nabla^2 H$. The Lagrange shift is applied in §6.

4 Tangential-tangential block

Under a purely tangential mode- m perturbation $\delta t_k = b \alpha_k$ with $\alpha_k = \cos(2\pi mk/N)$ (and $a = 0$ in (3)), the perturbation is $\delta z_k = i z_k b \alpha_k$, so $z_k(b) = z_k(1 + i b \alpha_k)$. A direct computation yields

$$|z_j(b) - z_k(b)|^2 = |z_j - z_k|^2 + 2b(\alpha_j - \alpha_k) \sin(\theta_j - \theta_k) + b^2 Q_{jk} + O(b^3), \quad (12)$$

with the same Q_{jk} as in (7). With $A = z_j - z_k$ and $B = \alpha_j z_j - \alpha_k z_k$,

$$|z_j(b) - z_k(b)|^2 = |A + ibB|^2 = |A|^2 - 2b \operatorname{Im}(\bar{A}B) + b^2 |B|^2,$$

and $\operatorname{Im}(\bar{A}B) = \operatorname{Im}[(z_j - z_k)^*(\alpha_j z_j - \alpha_k z_k)] = -(\alpha_j - \alpha_k) \sin(\theta_j - \theta_k)$ (from $\operatorname{Im}[\alpha_j + \alpha_k - \alpha_k e^{i(\theta_k - \theta_j)} - \alpha_j e^{i(\theta_j - \theta_k)}]$). The sign in (12) follows: $-2b \cdot [-(\alpha_j - \alpha_k) \sin(\theta_j - \theta_k)] = +2b(\alpha_j - \alpha_k) \sin(\theta_j - \theta_k)$.

Expanding $-\frac{1}{2} \ln |z_j(b) - z_k(b)|^2$ to second order in b (with $L_{jk} = 2(\alpha_j - \alpha_k) \sin(\theta_j - \theta_k)$ and $U_{jk} = |z_j - z_k|^2$),

$$-\ln |z_j(b) - z_k(b)| = -\ln |z_j - z_k| - \frac{bL_{jk}}{2U_{jk}} + b^2 \left[\frac{L_{jk}^2}{4U_{jk}^2} - \frac{Q_{jk}}{2U_{jk}} \right] + O(b^3).$$

The linear term, summed over pairs, vanishes: setting $p = k - j$ and pairing p with $N - p$, the factor $\sin(\theta_j - \theta_k) = -\sin(2\pi p/N)$ is odd in p while $(\alpha_j - \alpha_k)/U_{jk}$ is even in p , so $\sum_{j < k} L_{jk}/U_{jk} = 0$ term by term.

With $p = k - j \pmod{N}$ and $\theta_j - \theta_k = -2\pi p/N$,

$$U_{jk} = 4 \sin^2(\pi p/N), \quad L_{jk}^2 = 4(\alpha_j - \alpha_k)^2 \sin^2(2\pi p/N) = 4(\alpha_j - \alpha_k)^2 \cdot U_{jk} \cos^2(\pi p/N),$$

using $\sin^2(2\pi p/N) = 4 \sin^2(\pi p/N) \cos^2(\pi p/N)$. Therefore $L_{jk}^2/(4U_{jk}^2) = (\alpha_j - \alpha_k)^2 \cos^2(\pi p/N)/U_{jk}$.

Proceeding as in §3, applying $\sum_j (\alpha_j - \alpha_{j+p})^2 = N(1 - \cos(2\pi mp/N))$ and $\sum_j Q_{j,j+p} = N(1 - \cos(2\pi mp/N) \cos(2\pi p/N))$, the tangential-tangential block entry is

$$(\nabla^2 H)_{tt}^{(m)} = \sum_{p=1}^{N-1} \frac{-(1 - \cos(2\pi mp/N) \cos(2\pi p/N)) + 2(1 - \cos(2\pi mp/N)) \cos^2(\pi p/N)}{4 \sin^2(\pi p/N)}. \quad (13)$$

The bracketed numerator simplifies via $2 \cos^2(\pi p/N) = 1 + \cos(2\pi p/N)$:

$$\begin{aligned} & - (1 - \cos(2\pi mp/N) \cos(2\pi p/N)) + (1 - \cos(2\pi mp/N)) (1 + \cos(2\pi p/N)) \\ & = \cos(2\pi p/N) - \cos(2\pi mp/N). \end{aligned}$$

(The $\cos(2\pi mp/N) \cos(2\pi p/N)$ cross terms cancel, $-1 + 1 = 0$, and the remaining terms are as stated.)

Thus

$$(\nabla^2 H)_{tt}^{(m)} = \sum_{p=1}^{N-1} \frac{\cos(2\pi p/N) - \cos(2\pi mp/N)}{4 \sin^2(\pi p/N)} = C(1) - C(m), \quad (14)$$

where $C(k)$ is as in Corollary 1. Substituting,

$$(\nabla^2 H)_{tt}^{(m)} = \frac{m(N-m)}{2} - \frac{N-1}{2} = \frac{m(N-m) - (N-1)}{2}. \quad (15)$$

For $m \in \{1, \dots, N-1\}$, $m(N-m) \geq N-1$ with equality exactly at $m \in \{1, N-1\}$, so $(\nabla^2 H)_{tt}^{(m)} \geq 0$ with equality at the translation modes (consistent with the translation Goldstone structure: modes $m = 1$ and $m = N-1$ generate infinitesimal rigid translations, which lie in the kernel of $\nabla^2 H$).

5 Off-diagonal vanishing and trace identity

Lemma 3 (Off-diagonal vanishing). *For every $N \geq 2$ and every $m \in \{1, \dots, N-1\}$ the radial-tangential cross entry of $\nabla^2 H$ in the mode- m normalized Fourier basis vanishes:*

$$(\nabla^2 H)_{rt}^{(m)} = 0.$$

Proof. Under the combined perturbation $\delta z_k = z_k \alpha_k (a + ib)$ with $\alpha_k = \cos(2\pi mk/N)$, write $A = z_j - z_k$ and $B = \alpha_j z_j - \alpha_k z_k$. Then $z_j(a, b) - z_k(a, b) = A + (a + ib)B$, so

$$|z_j(a, b) - z_k(a, b)|^2 = |A|^2 + 2a \operatorname{Re}(A\bar{B}) + 2b \operatorname{Im}(A\bar{B}) + (a^2 + b^2) |B|^2.$$

Crucially, there is *no* ab term in $|v(a, b)|^2$ itself. The ab term in $-\frac{1}{2} \ln |v(a, b)|^2$ therefore comes entirely from the cross product of the two linear terms in the expansion $-\frac{1}{2} \ln(1+x) = -x/2 + x^2/4 - \dots$, namely

$$\text{coeff of } ab \text{ in } -\frac{1}{2} \ln |v(a, b)|^2 = \frac{P_{jk} R_{jk}}{2U_{jk}^2},$$

with $P_{jk} = 2 \operatorname{Re}(A\bar{B}) = 2(\alpha_j + \alpha_k)(1 - \cos(\theta_j - \theta_k))$ and $R_{jk} = 2 \operatorname{Im}(A\bar{B}) = 2(\alpha_j - \alpha_k) \sin(\theta_j - \theta_k)$. Multiplying and using $1 - \cos(\theta_j - \theta_k) = U_{jk}/2$,

$$\frac{P_{jk} R_{jk}}{2U_{jk}^2} = \frac{2(\alpha_j^2 - \alpha_k^2) \sin(\theta_j - \theta_k)}{U_{jk}}.$$

Summing over pairs with $p = k - j$ and fixing p , the dependence on j is only through $\alpha_j^2 - \alpha_{j+p}^2$. But $\sum_{j=0}^{N-1} \alpha_j^2 = N/2 = \sum_{j=0}^{N-1} \alpha_{j+p}^2$ (the second sum is a cyclic shift of the first), so

$$\sum_{j=0}^{N-1} (\alpha_j^2 - \alpha_{j+p}^2) = 0 \quad \text{for every } p \in \{1, \dots, N-1\}.$$

Therefore the ab coefficient in $H(a, b)$ vanishes, which (after unit normalization of $\hat{e}_r, \hat{e}_t$) gives $(\nabla^2 H)_{rt}^{(m)} = 0$. $\square$

Lemma 4 (Trace identity). $\lambda_m^+ + \lambda_m^- = N - 1$ for every $m \in \{1, \dots, N-1\}$.

Proof. Direct from (17): $[(N-1) - m(N-m)/2] + [m(N-m)/2] = N-1$. Equivalently, $\lambda_m^+ + \lambda_m^- = (\nabla^2 H)_{rr}^{(m)} + (\nabla^2 H)_{tt}^{(m)} + 2 \cdot (N-1)/2$, and the two unshifted sums telescope via (11) and (15): $(N-1 - m(N-m))/2 + (m(N-m) - (N-1))/2 = 0$, leaving $N-1$. $\square$

6 Lagrange shift and eigenvalues

The Lagrangian is $\mathcal{L} = H - \mu_L L$ with angular impulse $L = \sum_k |z_k|^2$ and $\mu_L = -(N-1)/4$ at the unit-ring equilibrium. Since $\nabla^2 L = 2I$, the Lagrangian Hessian is

$$\nabla^2 \mathcal{L} = \nabla^2 H - 2\mu_L I = \nabla^2 H + \frac{N-1}{2} I. \quad (16)$$

Adding $(N-1)/2$ to each of (11) and (15) yields

$$\lambda_m^+ = \frac{N-1 - m(N-m)}{2} + \frac{N-1}{2} = (N-1) - \frac{m(N-m)}{2}, \quad \lambda_m^- = \frac{m(N-m) - (N-1)}{2} + \frac{N-1}{2} = \frac{m(N-m)}{2} \quad (17)$$

The trace identity $\lambda_m^+ + \lambda_m^- = N-1$ falls out automatically — it is not an independent check but a consequence of the $(N-1)/2$ shift doubling across the two diagonal entries.